INCIDENT INVESTIGATION REPORT

IR-2025-017

P-101 Mechanical Seal Failure and Hydrocarbon Release

Incident Date:	January 15, 2025, 03:42 AM CST
Location:	Unit 100, P-101 Pump Skid, Petromax Baytown Complex
Severity Classification:	OSHA Recordable -- Near Miss (no injuries)
Investigation Lead:	T. Nakamura, Process Safety Engineer
Investigation Team:	R. Vasquez, D. Kim, M. Okonkwo, J. Rodriguez (Ops Supervisor)
Report Date:	February 28, 2025

1. Incident Description

On January 15, 2025 at approximately 03:42 AM CST, the Control Room Operator (CRO) received a high-level alarm on the P-101 mechanical seal barrier fluid reservoir (LIA-10105). The alarm indicated that the barrier fluid level had dropped below the low-low setpoint of 25%, suggesting a significant seal leak. Simultaneously, a combustible gas detector (GD-10103) located near the P-101 pump skid alarmed at 15% LEL.

The CRO immediately initiated the Emergency Operating Procedure EOP-100-01 (Unit 100 Pump Trip and Isolation). P-101 was tripped remotely from the control room at 03:44 AM. The suction and discharge block valves (HV-10101 and HV-10102) were remotely closed. The area fire water deluge system was placed in manual standby.

The on-shift Operations Supervisor (J. Rodriguez) dispatched two operators to the P-101 area at 03:50 AM. They confirmed visible crude oil spray from the outboard mechanical seal gland area. Estimated release volume: 5-8 gallons of crude oil, contained within the pump skid drip pan and secondary containment. No personnel were in the immediate vicinity at the time of the release. No fire or ignition occurred. The area was barricaded and the release was cleaned up by the environmental response team by 06:30 AM.

2. Timeline of Events

December 18, 2024, 14:00: Routine vibration survey by M. Okonkwo identifies slightly elevated readings at P-101 drive-end bearing (0.28 in/s vs. 0.18 in/s baseline). Noted as "watch" status in condition monitoring database.

January 8-10, 2025: Extreme cold weather event (temperatures below 10 degF for 72 hours). Plant operates in cold weather contingency mode. Suction piping to P-101 experiences partial trace heat failure; suction temperature drops from 250 degF to approximately 180 degF.

January 12, 2025, 08:00: Operators report unusual noise from P-101 during shift rounds. Noise described as intermittent "crackling" sound near the seal area. Noted in shift log but no work order raised.

January 13, 2025, 22:00: Barrier fluid consumption rate noted as increasing (0.3 liters/day, up from normal 0.05 liters/day). CRO noted in shift log.

January 15, 2025, 03:42: Barrier fluid low-low alarm. Gas detector alarm at 15% LEL. Pump tripped and isolated. Crude oil spray observed from seal area.

January 15, 2025, 06:30: Area cleaned, release contained. No environmental impact outside secondary containment.

3. Root Cause Analysis

The investigation team performed a root cause analysis using the TapRooT methodology and fault tree analysis. The physical root cause and contributing factors are summarized below.

3.1 Physical Root Cause

Examination of the removed mechanical seal (John Crane Type 4620, S/N: JC-46200-29847) by the manufacturer's failure analysis laboratory revealed the following:

- The outboard seal face (silicon carbide rotating element) exhibited severe thermal shock cracking. A radial crack approximately 7mm long propagated from the ID to the OD of the seal face.
- The inboard seal face showed heat checking patterns consistent with dry running.
- The elastomeric secondary seals (O-rings, Viton Grade GF) showed no degradation.

The thermal shock cracking was caused by rapid temperature transients during the January cold weather event. When suction temperature dropped from 250 degF to 180 degF over a short period, the differential thermal expansion between the silicon carbide seal face and the stainless steel seal hardware exceeded the design limits, initiating the crack.

3.2 Contributing Factors

1. HEAT TRACE FAILURE: The electric heat tracing on the P-101 suction piping (Circuit HT-101-S) failed due to a defective thermostat (Chromalox model 1900-11). This allowed the suction temperature to drop well below the minimum recommended pump inlet temperature of 220 degF.

2. INADEQUATE COLD WEATHER PROCEDURE: The existing cold weather contingency procedure (WI-CW-001) did not include specific guidance for monitoring P-101 suction temperature or triggering a pump trip on low suction temperature.

3. DELAYED RESPONSE TO ABNORMAL NOISE: The unusual noise reported on January 12 was an early indicator of seal distress. The observation was recorded in the shift log but was not escalated to the Reliability Engineer or Vibration Analyst for assessment. No work order was raised.

4. VIBRATION MONITORING GAP: The elevated vibration reading from December 2024 (0.28 in/s) was noted as "watch" status but the monitoring interval was not increased from monthly to weekly until after the January 22 follow-up survey -- which occurred AFTER the incident.

4. Corrective Actions

#	Action Item	Owner	Due Date	Status
1	Replace P-101 mech seal per SOP-MNT-001	D. Kim	04/15/25	COMPLETE
2	Repair heat trace circuit HT-101-S	Electrical	02/15/25	COMPLETE
3	Install low suction temp alarm on P-101	I&E	03/01/25	COMPLETE
4	Update cold weather procedure WI-CW-001	T. Nakamura	03/15/25	COMPLETE
5	Abnormal situation mgmt refresher training	Operations	04/30/25	COMPLETE
6	Review all pump seal barrier fluid alarms	I&E	05/30/25	IN PROGRESS
7	Evaluate upgrade to SiC/SiC seal faces	D. Kim	06/30/25	IN PROGRESS

5. Lessons Learned

1. ABNORMAL SITUATION RECOGNITION: Unusual equipment sounds are critical early warning indicators and must always be reported AND escalated to the appropriate technical resource (Reliability Engineer or Vibration Analyst). An "observe and report" culture alone is insufficient -- operators must have clear guidance on when to initiate a work order.

2. COLD WEATHER PREPAREDNESS: Heat tracing is a safety-critical system for process equipment operating above ambient temperature. Heat trace circuits for pump suction lines and seal systems should be verified functional before each winter season.

3. CONDITION MONITORING FOLLOW-UP: When vibration readings reach "watch" status, the monitoring frequency should be automatically escalated. This has been codified into the updated Vibration Monitoring Procedure (VIB-PROC-002, Rev. 5).

4. BARRIER FLUID MONITORING: The barrier fluid consumption rate is a leading indicator of seal health. An automatic trending alarm on consumption rate (not just level) should be implemented for all dual mechanical seal systems. This action is being tracked under Corrective Action #6.

6. PSM Classification

This incident is classified as a Near Miss under the Petromax Process Safety Management program. It is reportable to OSHA as a process safety incident under 29 CFR 1910.119 because it involved the release of a highly hazardous chemical (crude oil) above the threshold quantity. The incident has been entered into the Petromax SAFER incident tracking system (Case #PM-2025-00417) and will be presented at the Q2 2025 Plant Safety Review Meeting.